ErSE222: Machine learning in Geoscience

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Examples of generative modelling

- Can you imagine a **dog driving a car**?

- Can you imagine a **story of a young boy travelling to the moon**?

- Can you imagine a **soundtrack of a videogame**?

- Can you imagine a **panda surfing in the sea**?

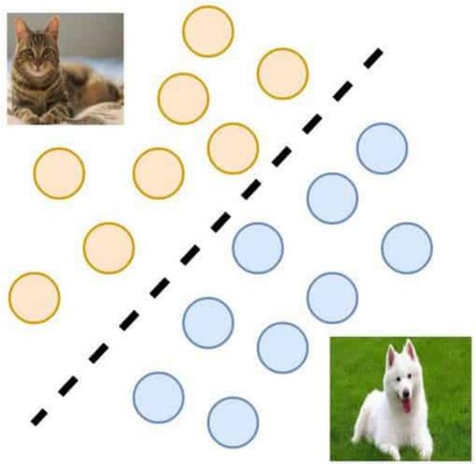
*As humans, we have **great imagination** -> and sometimes can put it into practice…*

Generative Modelling: what and how?

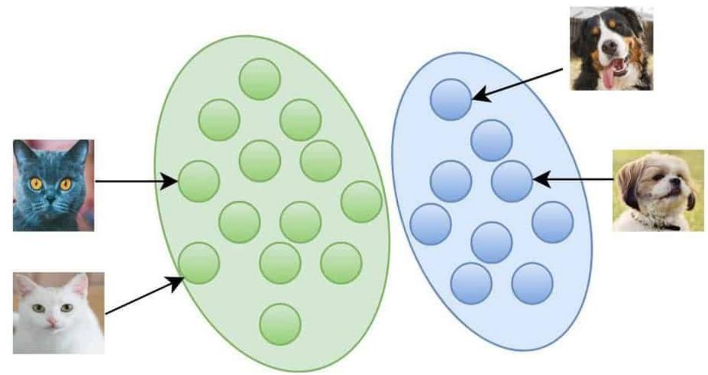
Generative modelling is a subset of artificial intelligence that aims to ***create new and original content*** *(images, text, music, videos, etc.)* ***using machine learning tools.***

- Some training data: $\mathbb{X} = \{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, ..., \mathbf{x}^{(N_s)}\}$
- A model with learnable parameters (which learns the data): f_θ
- A procedure for the model to learn: $\mathbf{x}^{(i)} \longrightarrow f_\theta$
- A procedure to generate data (or evaluate prob): $f_\theta \rightarrow \mathbf{x}^{(new)}/p(\mathbf{x}^{(new)})$

Generative Modelling: what and how?



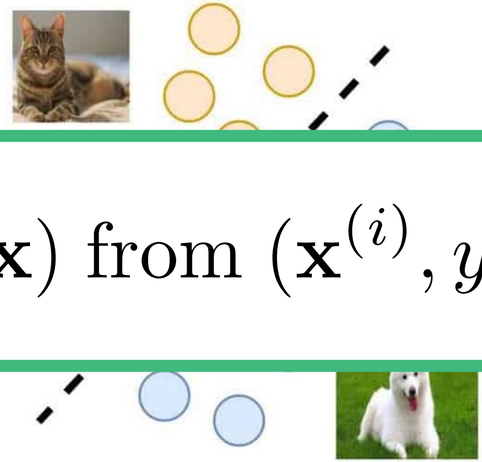
Discriminative



Generative

Source: https://learnopencv.com/generative-and-discriminative-models/

Generative Modelling: what and how?



$$p(y|\mathbf{x}) \text{ from } (\mathbf{x}^{(i)}, y^{(i)})$$

Discriminative

$$p(\mathbf{x}) \text{ from } \mathbf{x}^{(i)}$$

or $p(\mathbf{x}|y)$ from $(\mathbf{x}^{(i)}, y^{(i)})$

Generative

Source: https://learnopencv.com/generative-and-discriminative-models/

Generative Modelling: what and how?

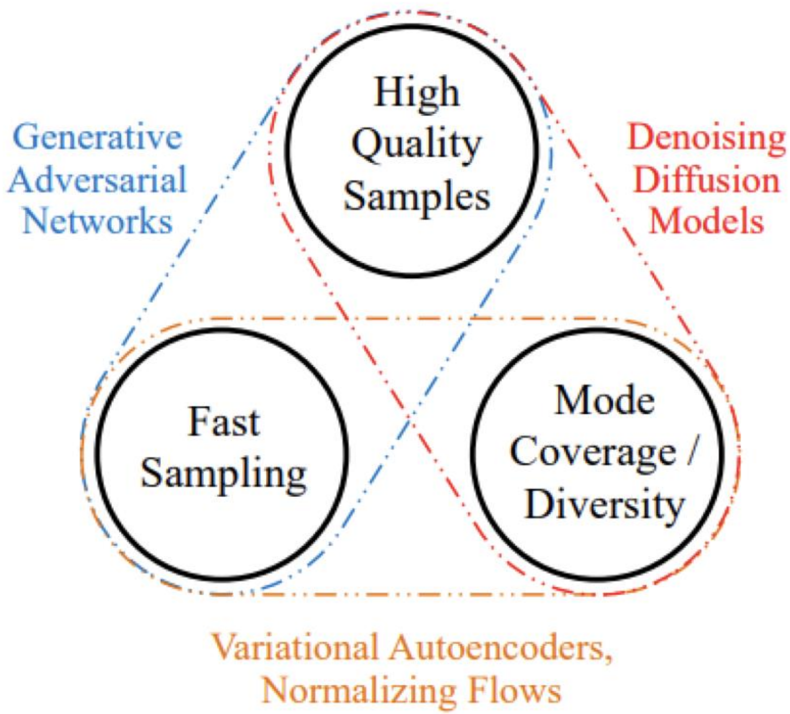
Ideally a generative modelling is a model solving the following problem

$$\underset{\theta}{\operatorname{argmax}}\ p(\mathbf{x}) \text{ from } \mathbf{x}^{(i)}$$

Maximum-likelihood estimator

We will see that this is not easy to do, usually approximations are required.

Generative Modelling: types of models

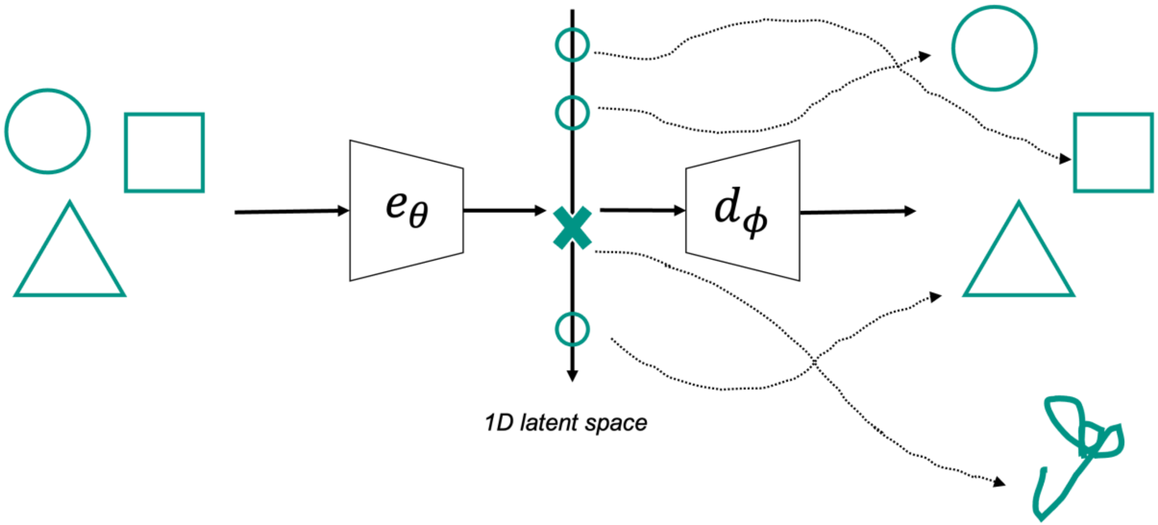


Generative learning trilemma

From: Tackling the Generative Learning Trilemma with Denoising Diffusion GANs

AutoEncoders as generative models

What if we **sample a new embedding vector** from z and then have the decoder reconstruct the “input” from it?



*Latent space is too **discontinous***

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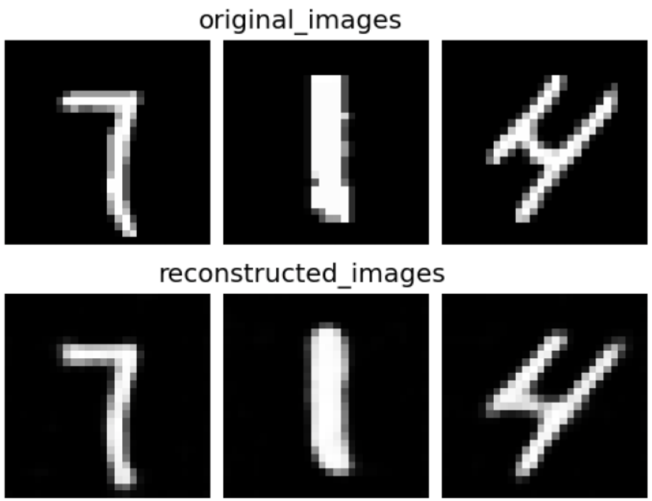
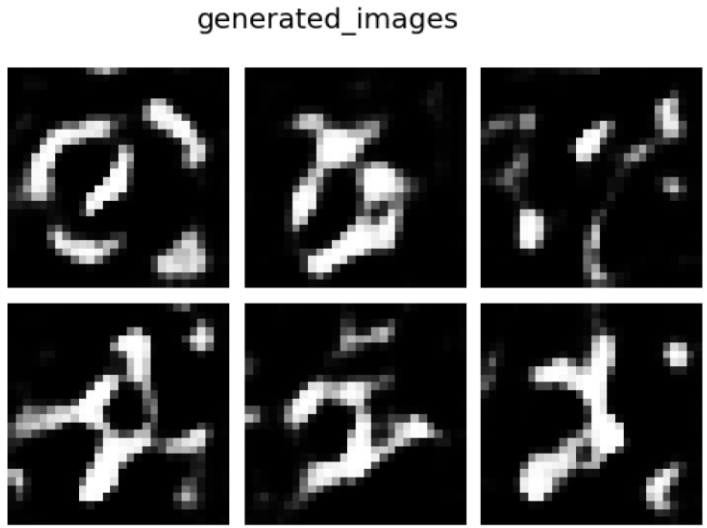
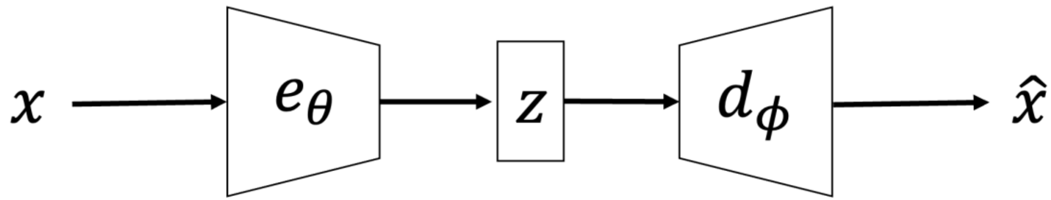




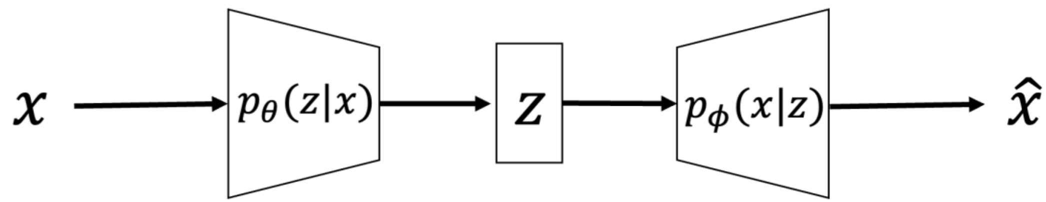
Latent space is too ***discontinous***

Variational AutoEncoders

AE



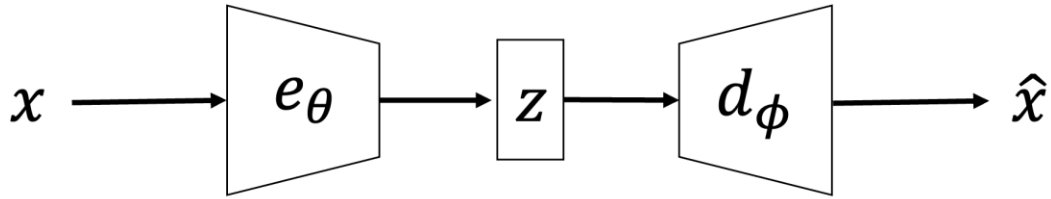
VAE



Add a probabilistic spin to AE: allow ***sampling the latent space***

Variational AutoEncoders

AE



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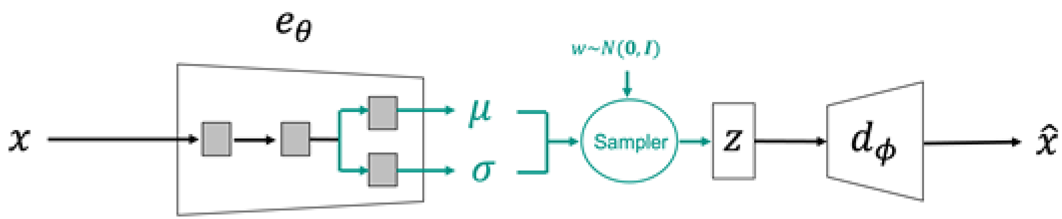
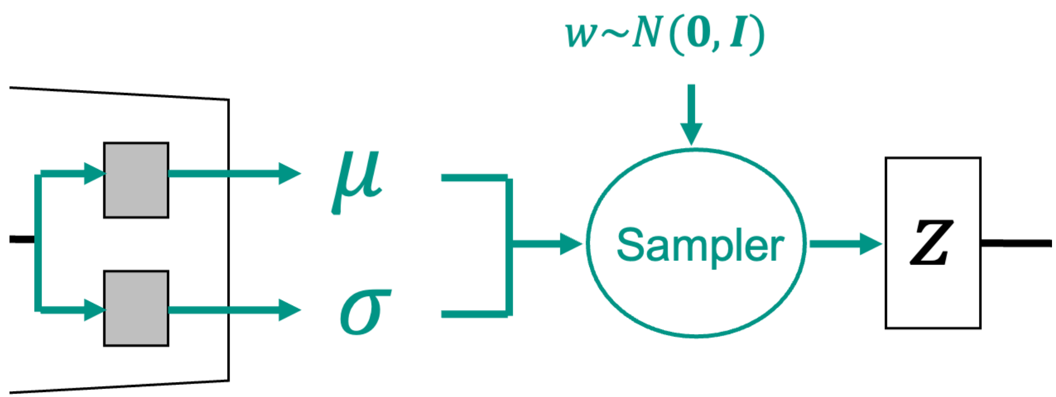


Add a probabilistic spin to AE: allow ***sampling the latent space***

Variational AutoEncoders

Modification 1: encoding process

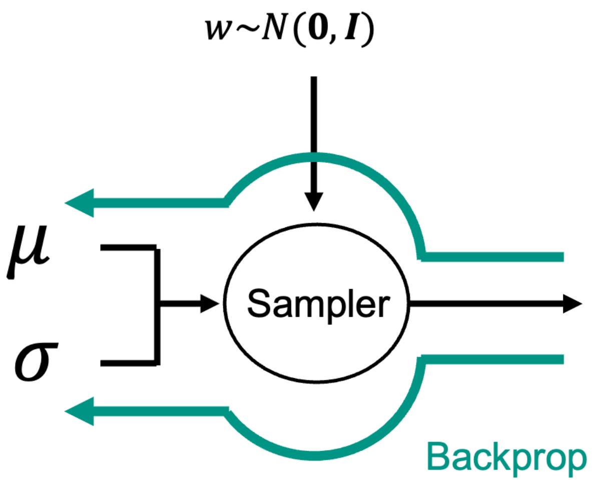


$q_\theta(\mathbf{z}) \approx p(\mathbf{z}|\mathbf{x})$: Proposal distribution

$p(\mathbf{z})$: Prior distribution

The Reparametrization trick

Way to sample a distribution whilst still being able to do backprop:



$$z = \mu + \sigma \cdot w$$

$$\frac{\partial z}{\partial \mu} = 1$$

$$\frac{\partial z}{\partial \sigma} = w$$

Variational AutoEncoders

Modification 2: loss function

$$\mathcal{L}_r = \frac{1}{N_s}\sum_i \Big(\mathcal{L}(\mathbf{x}^{(i)}, d_\phi(e_\theta(\mathbf{x}^{(i)}))) + KL(p(\mathbf{z}^{(i)}|\mathbf{x}^{(i)})||p(\mathbf{z}^{(i)}))\Big)$$

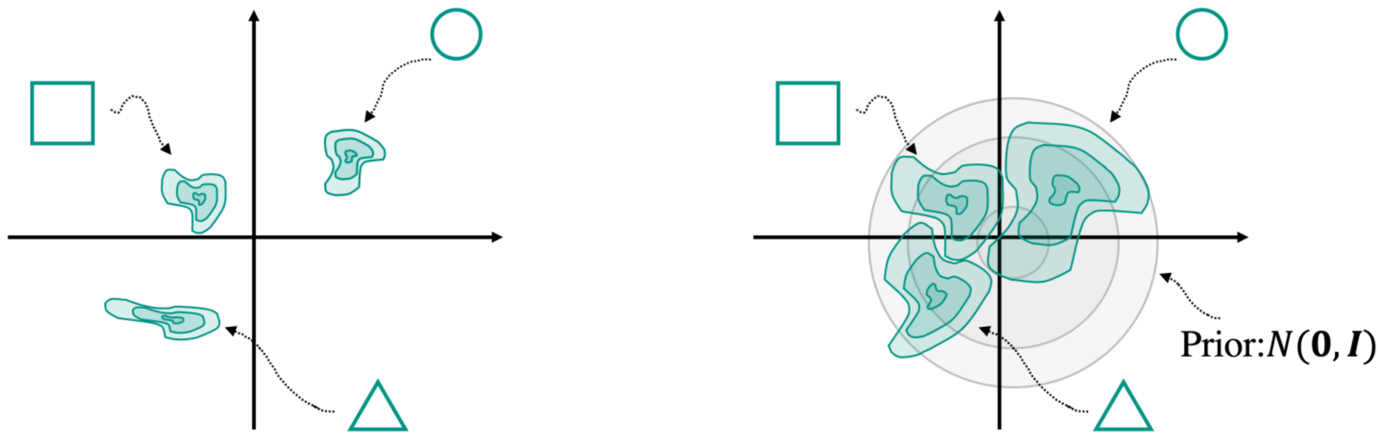
Reconstuction Loss (as in AEs) **Kullback-Leibler divergence**

Comes from solving this objective function (see notes for derivation):

$$\underset{\theta}{\operatorname{argmin}}\; KL(q_\theta(\mathbf{z})||p(\mathbf{z}|\mathbf{x}))$$

Variational AutoEncoders

<u>Modification 2</u>: loss function



AE latent space

VAE latent space

VAEs in practice

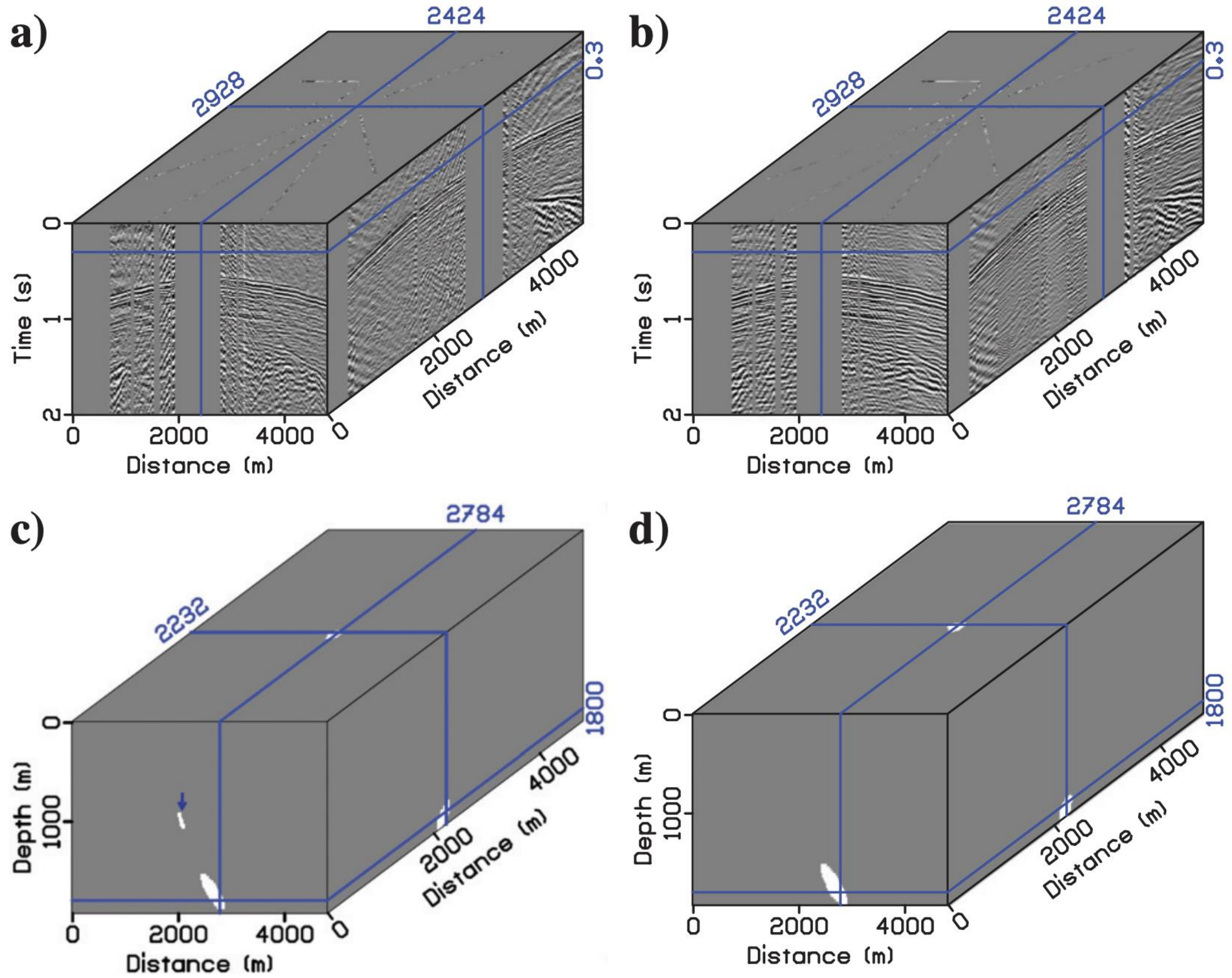
Learning Algorithm

1. Select a batch of training samples and feed to encoder
2. Divide the output of the encoder into two parts: mean and std (each of them has the same number of samples as the input batch)
3. Sample a latent space realization per training sample
4. Feed each latent vector into the decoder
5. Compute the loss: 1) MSE of reconstruction, 2) KL between prior and proposal using the mean and std obtained in step2. + backprop.

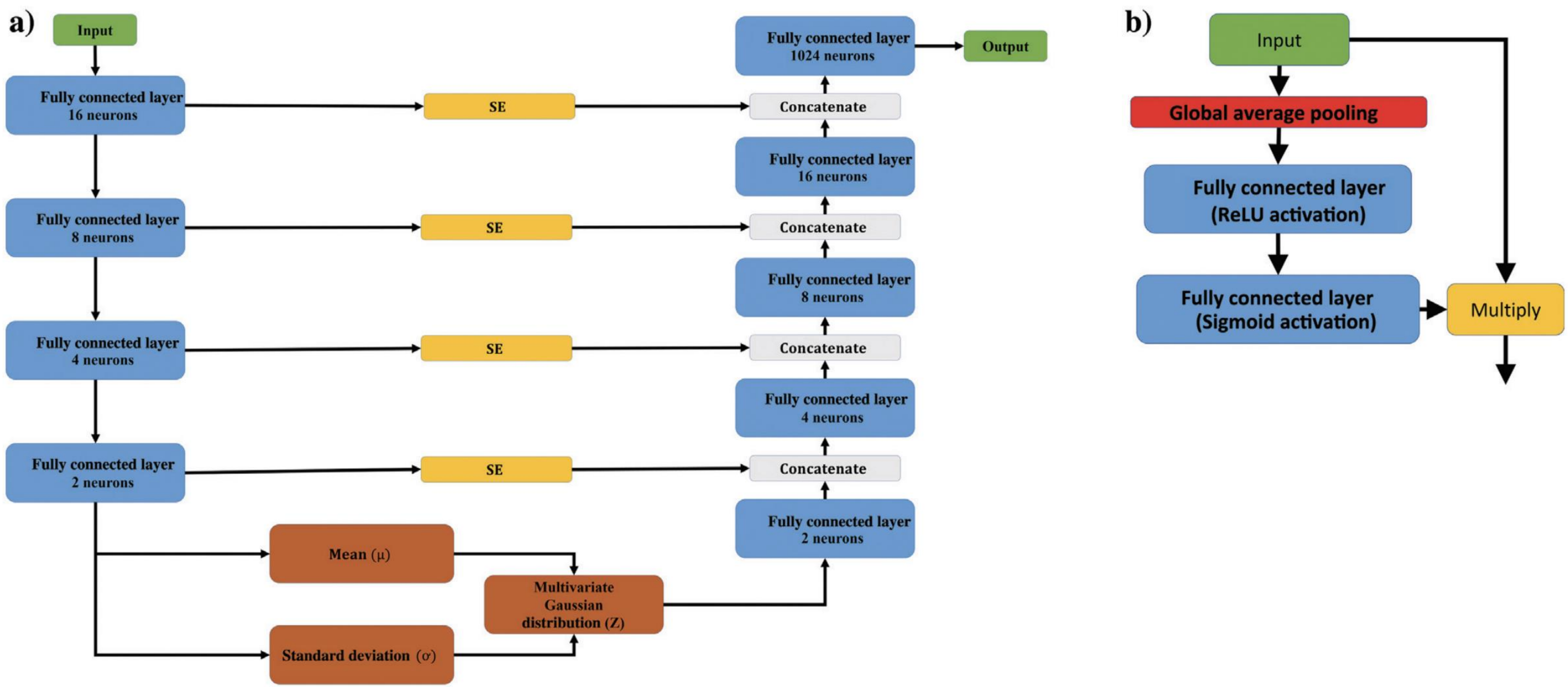
Sampling Algorithm

1. Sample a latent space realization
2. Feed each latent vector into the decoder

Applications



Uncovering the microseismic signals from noisy data for high-fidelity 3D source-location imaging using deep learning



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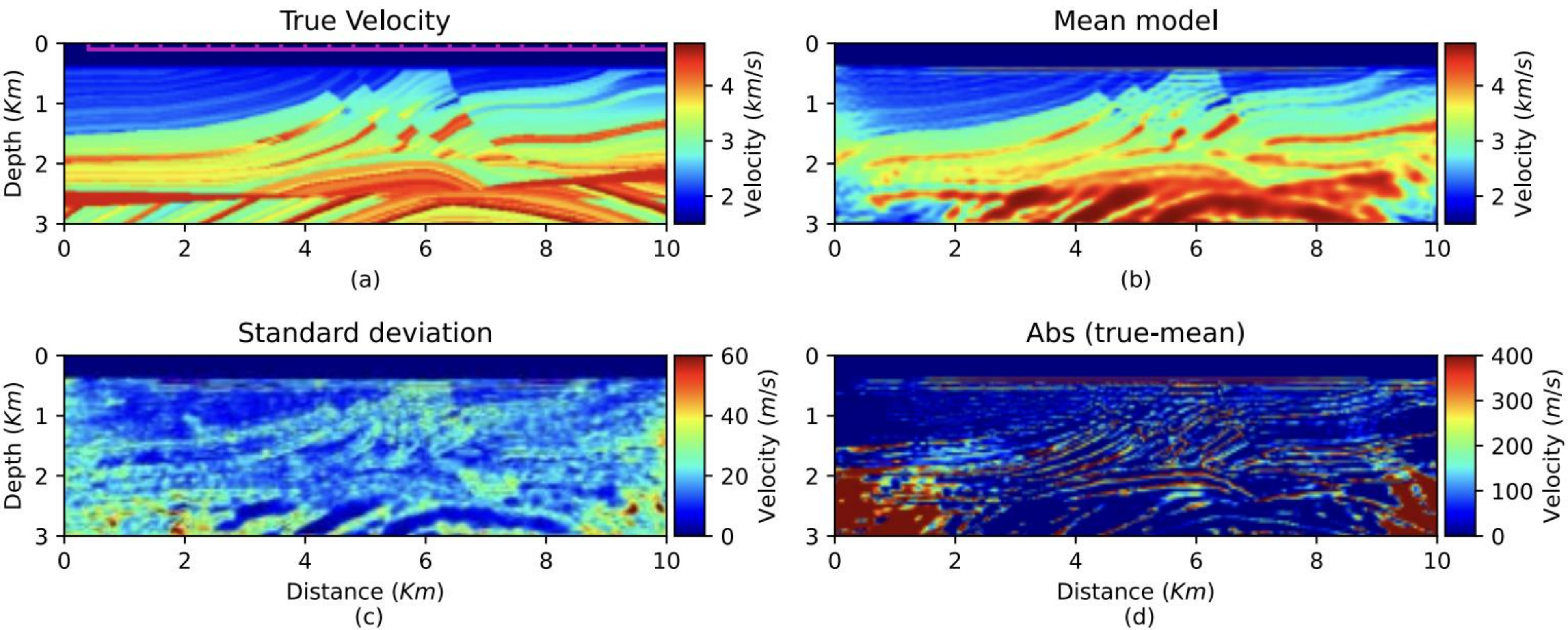
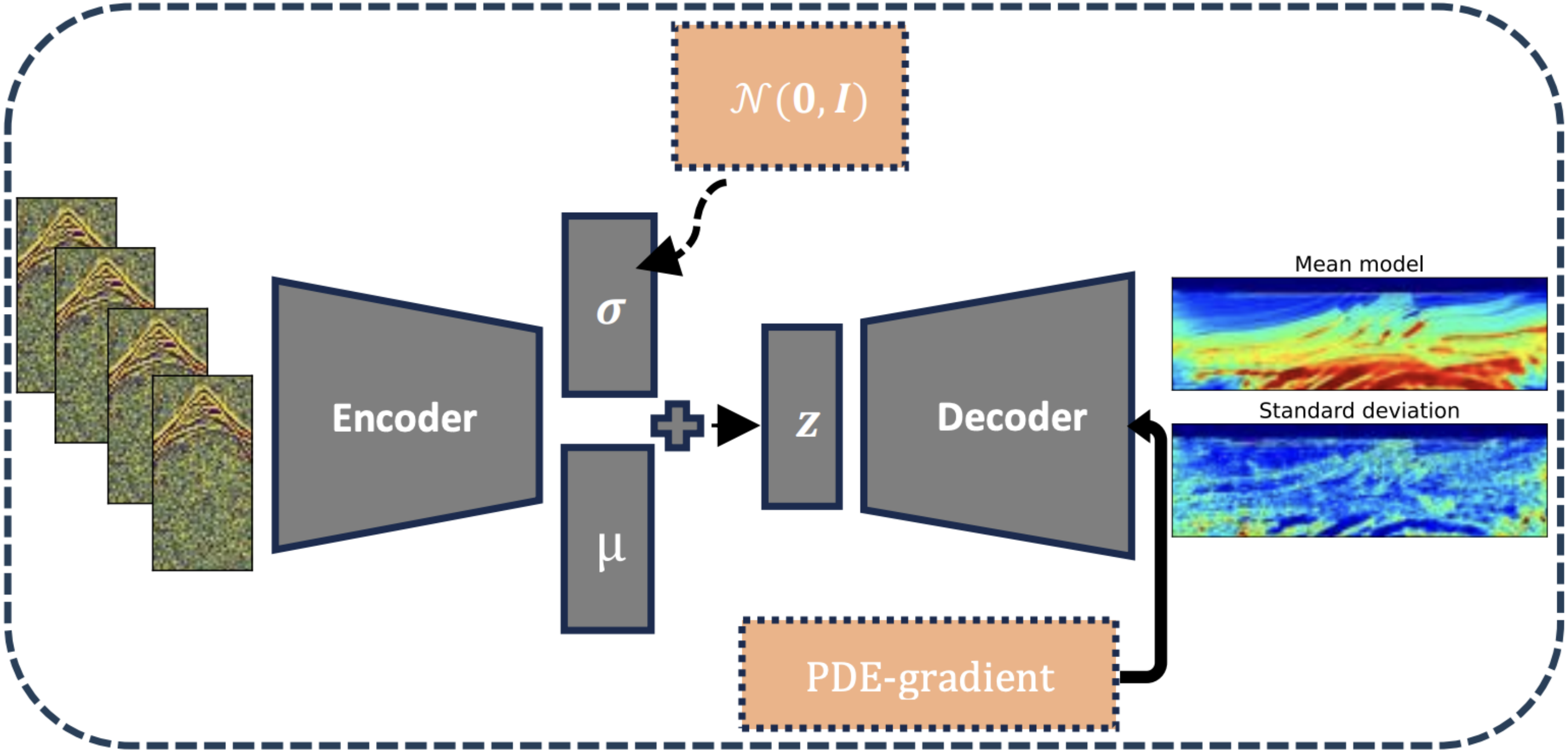


Uncertainty quantification of single and multi-parameter full-waveform inversion through a variational autoencoder



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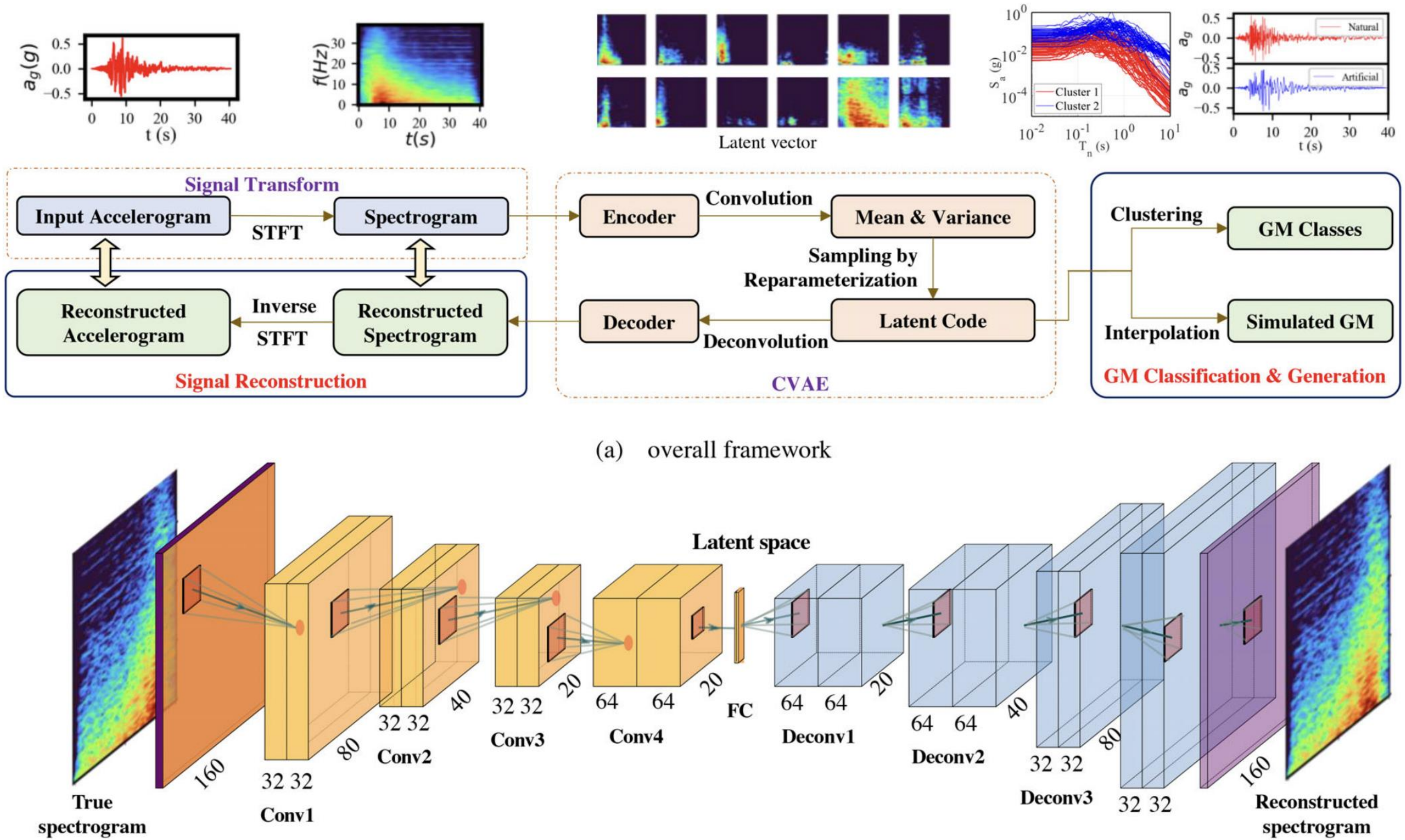

(a) overall framework

Convolutional variational autoencoder for ground motion classification and generation toward efficient seismic fragility assessment